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Large deformation isotropic elasticity: on the correlation of theory and experiment for compressible rubberlike solids

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A method of approach to the correlation of theory and experiment for incompressible isotropic elastic solids under finite strain was developed in a previous paper (Ogden 1972). Here, the results of that work are extended to incorporate the effects of compressibility (under isothermal conditions). The strain-energy function constructed for incompressible materials is augmented by a function of the density ratio with the result that experimental data on the compressibility of rubberlike materials are adequately accounted for. At the same time the good fit of the strain-energy function arising in the incompressibility theory to the data in simple tension, pure shear and equibiaxial tension is maintained in the compressible theory without any change in the values of the material constants.

A full discussion of inequalities which may reasonably be imposed upon the material parameters occurring in the compressible theory is included.

1. INTRODUCTION

In a recent paper (1972, henceforth referred to as (I)) Ogden has developed a novel approach to the correlation between theory and experiment for the isothermal mechanical behaviour of *incompressible* isotropic elastic solids. In (I) the finite deformation characteristics of rubberlike solids are considered in relation to the theory. These materials are effectively incompressible in most situations and may also be regarded as isotropic relative to an undistorted stress-free ground state (which we assume to exist). The basis of the theory lies in the expansion of the strain-energy function W (per unit volume in the ground state) as a symmetric function of the three principal components of stretch a_1, a_2 and a_3 subject to the incompressibility constraint $a_1 a_2 a_3 = 1$. This differs from the customary approach in which W is expanded as a function of a particular pair of independent stretch invariants. In the development of the present method the use of analytical techniques associated with the principal axes of stress and strain is essential. These techniques, a study of which was initiated by Hill (1970*a, b*), have proved to be very useful in problems where the deformations are not as simple as those considered in (I), and more can be learned than by the conventional approach (see Ogden & Chadwick 1972; Ogden, Chadwick & Haddon 1972; Chadwick & Haddon 1972).

For incompressible materials $W(a_1, a_2, a_3)$ can be expanded on any closed interval in terms of linear combinations of expressions of the form

$$a_1^\alpha (a_2^\beta + a_3^\beta) + a_2^\alpha (a_3^\beta + a_1^\beta) + a_3^\alpha (a_1^\beta + a_2^\beta) - 6, \quad (1)$$

[567]

where α and β are real numbers. These are symmetric with respect to interchange of any pair of a_1 , a_2 and a_3 . In a formal expansion α and β would normally be integers, but this is not an essential requirement. In (I) the expression (1) is simplified by setting $\beta = 0$ and the possibility of α taking non-integral (real) values is allowed; a very good correlation between theory and experiment is then obtained by use of a strain-energy function consisting of a linear combination of terms like

$$\phi(\alpha) \equiv (a_1^\alpha + a_2^\alpha + a_3^\alpha - 3)/\alpha. \quad (2)$$

It may be written

$$W = \mu_n \phi(\alpha_n), \quad (3)$$

where summation over a number (for the moment unspecified) of terms is implied by the repetition of the suffix n . For the particular rubber considered in (I) three (α_n, μ_n) pairs are required and the values of the α_n obtained are 1.3, 5.0 and -2.0 . The corresponding values of the μ_n are 6.3, 0.012 and -0.1 kgf cm^{-2} (10^5 Pa) respectively.† The use of this explicit form of strain-energy function has led to numerical predictions of the behaviour of rubberlike solids in various situations, some of which have not yet been fully explored experimentally (see Ogden & Chadwick 1972; Ogden *et al.* 1972). It should be remarked that the invariants I_1 and I_2 which are normally used in the polynomial expansion for W are given by $I_1 = 2\phi(2) + 3$ and $I_2 = -2\phi(-2) + 3$, for incompressible materials.

For many purposes the model of incompressibility leads to an adequate theoretical representation of the mechanical response of rubberlike materials. However, all real materials are compressible in some degree and they also admit longitudinal waves with a finite speed of propagation. Moreover, the idea of incompressibility has not been incorporated successfully into any adequate thermodynamic theory of rubber elasticity. It is therefore thought desirable to extend the results obtained in (I) to allow for volume changes. We do this in the present paper, following the approach initiated in (I).

It is appropriate to make brief mention here of the experimental work relating to the compressibility of rubberlike materials which is relevant to the present investigation. Bridgman (1944) measured the isothermal volume changes in 14 different rubbers subject to pressures of up to $25\,000 \text{ kgf cm}^{-2}$ (2.5 GPa) and obtained reductions in volume of as much as 20 %. Similar experiments had been carried out previously by Adams & Gibson (1930) at pressures of up to $12\,000 \text{ kgf cm}^{-2}$ (1.2 GPa). Subsequently, Weir (1953) performed experiments over a range of pressures equivalent to that used by Adams & Gibson, but for a series of temperatures ranging from 10 to 81.5°C . Other authors (e.g. Scott 1935; Copeland 1948; Wood & Martin 1964) have considered the effect of *thermal* expansivity also. However, we are not concerned with thermal effects in this paper.

Gee (1946) predicted theoretically that in simple tension the volume would increase slightly at small extensions (i.e. up to about 100 % extension) and that at higher extensions (above about 400 %) would decrease, this latter effect being

† $1 \text{ kgf} \approx 9.81 \text{ N}$; $1 \text{ kgf cm}^{-2} \approx 9.81 \times 10^4 \text{ Pa}$.

due to crystallization. The first of these predictions was confirmed by the careful experiments of Gee, Stern & Treloar (1950) and Hewitt & Anthony (1958). The second prediction had previously been shown to be true by Holt & McPherson (1936) who demonstrated that after about 400 % extension the volume decreases, in fact by as much as 2 % at 700 % extension. Their experiments, however, were not accurate enough to show the initial expansion effect because this is of the same order as their experimental error. Some data of the above-mentioned experimenters will be considered in § 3.

In § 2 we generalize the strain-energy function (3) to allow for volume change effects; these are accounted for by the addition of a function, $F(a_1 a_2 a_3)$, of $a_1 a_2 a_3$ to the form (3). Some particular deformations are then considered and the predictions of the compressible and incompressible theories in corresponding situations are compared. In particular, it is noted that the admission of compressibility has no significant effect on the force-stretch curves in simple tension, pure shear and equibiaxial tension obtained in (I) for the incompressible theory provided the constants α_n and μ_n are unchanged in value. The present theory is specialized to the case of infinitesimal deformations and related to the classical theory by defining, in an appropriate manner, the material constants occurring in W .

A simple form of F is considered in § 3 and successfully related to some experimental data of Bridgman (1944) and other authors on the compressibility of rubberlike materials. Use is made of the fact that the bulk modulus for rubberlike materials is several orders of magnitude larger than the shear modulus.

Finally, in § 4, we discuss what restrictions might reasonably be imposed on the material constants α_n and μ_n and on the function F . Extending the discussion given in ((I), § 6) for incompressible materials, we give sufficient conditions for Hill's (1968, 1970*b*) constitutive inequality to be satisfied and then consider some consequences of these conditions. In particular, it is shown that they necessitate a volume expansion of the correct order of magnitude in simple tension.

2. A STRAIN-ENERGY FUNCTION FOR COMPRESSIBLE RUBBERLIKE MATERIALS

In § 1 it was pointed out that for incompressible isotropic elastic solids the strain-energy function can be expanded in terms of the expressions (1). Correspondingly the expansion for compressible materials may be put in terms of the invariant quantities

$$a_1^\alpha(a_2^\beta a_3^\gamma + a_2^\gamma a_3^\beta) + a_2^\alpha(a_3^\beta a_1^\gamma + a_3^\gamma a_1^\beta) + a_3^\alpha(a_1^\beta a_2^\gamma + a_1^\gamma a_2^\beta) - 6$$

or, equivalently,

$$\{a_1^\alpha(a_2^\beta + a_3^\beta) + a_2^\alpha(a_3^\beta + a_1^\beta) + a_3^\alpha(a_1^\beta + a_2^\beta)\}(a_1 a_2 a_3)^\gamma - 6, \quad (4)$$

where α , β and γ are real numbers. For compressible materials the strain-energy function W has usually been expressed as a function of three independent stretch invariants, most frequently $I_1 = a_1^2 + a_2^2 + a_3^2$, $I_2 = a_2^2 a_3^2 + a_3^2 a_1^2 + a_1^2 a_2^2$ (or I_2/I_3) and

$I_3 = a_1^2 a_2^2 a_3^2$; but, as observed in (I), the expansion for W as a polynomial in I_1 , I_2 and I_3 can lead to unnecessary algebraic complexity. The approach adopted in (I) and continued here is to regard the principal stretches themselves as independent variables and, where necessary, to use techniques for dealing with the associated principal axes of stress and strain. The primary objective, as in (I), is to find a strain-energy function which, while being of a relatively simple form, predicts the actual mechanical behaviour of rubberlike solids as accurately as required.

Our starting-point is the set of invariant forms (4). In (I) it was found that, by use of a strain-energy function of the form (3) and with the assumption of incompressibility, a good correlation between theory and the experimental data in simple tension, pure shear and equibiaxial tension could be obtained. The term $\phi(\alpha)$, defined by equation (2), is obtainable from (4) with $\beta = \gamma = 0$. In order not to interfere with the good fit of the form (3) to the experimental data in circumstances where volume changes are of a small order (as they are for the deformations considered in (I); see (a) below) we retain in the expansion for the compressible form of W the terms corresponding to $\beta = \gamma = 0$ from (4); (note that a_1 , a_2 and a_3 no longer satisfy the identity $a_1 a_2 a_3 \equiv 1$). In addition, we include terms corresponding to $\alpha = \beta = 0$ in (4) as a measure of the volume change effects.

For compressible materials we therefore write

$$W = \mu_n \phi(\alpha_n) + F(a_1 a_2 a_3), \quad (5)$$

where F is a function of the density ratio $\rho_0/\rho \equiv a_1 a_2 a_3$ such that $F(1) = 0$; hence, in the incompressible limit, (5) reduces to (3). The densities ρ_0 and ρ refer to the ground state and current configuration respectively. For analytical purposes the general form (5) is retained, neither the number of the constants α_n and μ_n nor their values being specified.

The strain-energy function (5) is a considerable reduction of the most general expansion in terms of the expressions (4). However, nothing is likely to be gained in the present context by a consideration of forms more general than (5); indeed, such forms would introduce a mathematical complexity greater than our objectives could justify. By way of justification for the adoption of the particular strain-energy function (5), the following remarks are called for. First, it should be noted that the term $F(a_1 a_2 a_3)$ does not enter into the calculation of the force-stretch curves for simple tension, pure shear and equibiaxial tension (recall (I), equations (15), (16) and (17)). Experimental evidence indicates that volume changes are of a small order in these stress systems and, therefore, the consequences of (5) will be indistinguishable from those of (3) in such situations. Furthermore, the small volume changes which do occur are accounted for via the terms $\mu_n \phi(\alpha_n)$ by virtue of the fact that $a_1 a_2 a_3$ differs from unity by a small quantity. The foregoing points are discussed in detail in subsection (a) below.

Secondly, the bulk modulus (according to any reasonable definition) of elasticity for rubberlike materials is several orders of magnitude larger than the ground-state shear modulus μ ; and the contributions to the stress components from the

$\mu_n \phi(\alpha_n)$ terms are of order μ (for sensible values of the a_i). Thus, it is reasonable to suppose that any additional terms should contribute only a hydrostatic part to the stress since additional non-hydrostatic terms would have significant effects on the simple tension and other equations which already account for the appropriate data satisfactorily. Furthermore, it is likely that 'large' volume changes will be achieved only in situations where the stress is one of all-round pressure or tension (or approximately so). On these grounds it appears that there is no need to include in the strain-energy function invariants (4) for which either $\gamma = 0$, $\alpha \neq 0$, $\beta \neq 0$ or $\alpha \neq 0$ (or $\beta \neq 0$). The compressibility is therefore incorporated into the strain-energy function through the function F of the (local) volume ratio $a_1 a_2 a_3$ (henceforth we denote $a_1 a_2 a_3$ by J). And F , which involves some constant (λ , say) with $\lambda \gg \mu$, contributes a hydrostatic part to the stress. This is the analogue of the 'arbitrary hydrostatic pressure' occurring in the incompressible theory and, in the limit $a_1 a_2 a_3 \equiv 1$, takes on this role.

The stress-stretch relations resulting from (5) are

$$\tau_i \equiv J \sigma_i = a_i \partial W / \partial a_i = \mu_n \alpha_i^n + J F'(J) \quad (i = 1, 2, 3), \quad (6)$$

where τ_i and σ_i are respectively the principal components of the Kirchhoff and Cauchy stress tensors, and $F' \equiv dF/dJ$. This compares with the relation

$$\sigma_i = \mu_n \alpha_i^n - p \quad (i = 1, 2, 3) \quad (7)$$

from the incompressible theory ((I), equation (14)). In equation (7) p is a hydrostatic stress not determined by the deformation.

At this point it is instructive to relate the equations (6) to those occurring in the classical linearized theory of isotropic elasticity. We set $a_i = 1 + e_i$, where e_i ($i = 1, 2, 3$) are the principal components of infinitesimal strain such that $|e_i| \ll 1$. Putting $e_1 + e_2 + e_3 = \epsilon$, we obtain $J = 1 + \epsilon$ to the first order in small quantities; also, expanding (6), we get

$$\tau_i = \mu_n \alpha_n e_i + \{F'(1) + F''(1)\} \epsilon + F'(1) + \sum_n \mu_n \quad (i = 1, 2, 3), \quad (8)$$

to the same order. It should be understood that, in the first term on the right-hand side of equation (8), summation is implied by the occurrence of the repeated suffix n .

We now assume that the unstrained state is stress free and therefore introduce the function $f(J)$, defined by

$$f(J) = F(J) + \left(\sum_n \mu_n \right) \ln J,$$

with the requirement $f'(1) = 0$. Then, to the first order, equation (8) takes on the classical form $\sigma_i \equiv \tau_i = \mu_n \alpha_n e_i + f''(1) \epsilon$, the Lamé constants μ (the shear modulus in the ground state) and λ being given by $2\mu = \mu_n \alpha_n$ (summed over n) and $\lambda = f''(1)$. It is convenient to introduce a new function $g(J)$ defined by $f = \lambda g$; then

$$g(1) = g'(1) = 0, \quad g''(1) = 1. \quad (9)$$

The strain-energy function (5) may now be expressed

$$W = \mu_n \{ \phi(\alpha_n) - \ln J \} + \lambda g, \quad (10)$$

the dependence of g upon J being understood, and equations (6) may be rewritten as

$$\tau_i = \mu_n (a_i^{2n} - 1) + \lambda J g' \quad (i = 1, 2, 3). \quad (11)$$

It is emphasized that in (11) and henceforth the appearance of the repeated suffix n implies summation over a number of terms unless an explicit statement is made to the contrary.

A further correspondence with the classical theory may be asserted by the introduction of Poisson's ratio ν , having the value $\frac{1}{2}$ in the incompressible limit, through the relation $\lambda = 2\mu\nu/(1-2\nu)$. This relation between the material constants μ , λ and ν is precisely that occurring in the classical theory. By direct analogy with classical theory, Blatz & Ko (1962) and Blatz (1963) defined a finite strain Poisson's ratio, ν^* say, as minus the ratio of lateral (logarithmic) strain to longitudinal (logarithmic) strain in simple tension. For the particular rubber and range of strain (up to stretch ratios of about 1.6) they considered, ν^* can be regarded as a material constant. In general, though, ν^* is strain dependent. It seems desirable, given a particular constitutive relation, that mechanical characteristics be specified by means of (strain-independent) *constants* (such as ν) which can be determined explicitly for any given rubber. In this paper it is convenient to work in terms of the constant λ rather than ν .

(a) *Some physical considerations*

We have already stated that the bulk modulus is very much larger than the shear modulus for rubberlike solids (in fact, typically by a factor of about 10^4). It follows that a 'large' volume reduction from the ground state can be achieved only by a pressure whose magnitude is much greater than that of the shear modulus. Use is made of this observation in the following physical discussion.

Consider a configuration of pure hydrostatic pressure P (> 0) in which the three principal stretches have the common value a and the all-round Kirchhoff stress is τ . Then $a = J^{\frac{1}{3}}$, and the pressure-density relation obtained from equation (11) is

$$-JP \equiv \tau = \mu_n (J^{\frac{2}{3}n} - 1) + \lambda J g'. \quad (12)$$

Provided g' is a well-behaved function (which we assume to be the case) it can be regarded as being of order unity, and the ratio of the terms $\mu_n (J^{\frac{2}{3}n} - 1)$ and $\lambda J g'$ is then typically of order μ/λ . The constant λ is a measure of the bulk modulus of elasticity ($\gg \mu$) of the considered rubber. Equation (12) is therefore able to predict values of P of the required order of magnitude. Since $g'(1) = 0$, it is clear that the term $\lambda J g'$ must be negative for $J < 1$. Without loss of generality we may take $\lambda > 0$; then we require $g' < 0$ for $J < 1$.

Little has been determined experimentally about situations in which $J > 1$. One would expect, however, that a correspondingly large all-round *tension* is required

to achieve volume *expansion* from the ground state. Therefore, the derivative of the function g should conform with the restrictions

$$\left. \begin{aligned} g' &> 0 & \text{for } J > 1, \\ g' &< 0 & \text{for } J < 1, \end{aligned} \right\} \quad (13)$$

in addition to the conditions (9). For $J \leq 1$ it is also reasonable to suppose that the pressure is a monotonic increasing function of the density; in other words, that g' is a monotonic increasing function of J for $0 < J \leq 1$. In such circumstances the second derivative g'' is positive for these values of J . Can the same restriction be expected for $J > 1$? It seems likely (though it is not certain) that, so long as the material remains elastic, an *increase* in all-round tension would always be required to effect a volume *increase*; we would then have $g'' > 0$ for all J (> 0). Once the 'elastic limit' is reached the possibility of g'' being less than zero cannot be ruled out. Further consideration will be given to inequalities involving the functions g' and g'' in § 4.

In this paper we shall not consider J being greater than unity when (in § 3) we relate particular forms of g to experimental data. The only situation in which we consider volume expansion is simple tension where the volume increases by an amount of order 10^{-4} for 'small' extensions. If the need arises at any future date the function g can be specialized to allow for other volume expansion effects.

Consider now a configuration of simple tension where the principal component τ_1 of Kirchhoff stress in the direction of extension is denoted by τ . The lateral stress components τ_2 and τ_3 are zero. The extension ratio is denoted by a so that $a_2 = a_3 = J^{\frac{1}{2}}a^{-\frac{1}{2}}$, and the resulting stress-stretch relations are, from equations (11),

$$\tau = \mu_n(a^{\alpha_n} - 1) + \lambda J g', \quad (14)$$

$$0 = \mu_n(J^{\frac{1}{2}}a^{\alpha_n} - 1) + \lambda J g', \quad (15)$$

(compare with the corresponding equations in the incompressible theory; see (I), § 3(a)). The resulting force-stretch relation is then, on elimination of the hydrostatic part between equations (14) and (15),

$$\tau/a = \mu_n\{a^{\alpha_n-1} - a^{-(1+\frac{1}{2}\alpha_n)}J^{\frac{1}{2}\alpha_n}\}. \quad (16)$$

The corresponding relation arising in the incompressible theory ((I), equation (15)) is

$$\sigma/a = \mu_n\{a^{\alpha_n-1} - a^{-(1+\frac{1}{2}\alpha_n)}\}. \quad (17)$$

As we noted in § 1, experiment indicates that J differs from unity by only a small amount ($\approx 10^{-4}$) for extensions up to about 400 %. Therefore, the force-stretch relation calculated by means of equation (16) will be indistinguishable from that calculated via (17) if the same values of α_n and μ_n are used in the two theories. The same remarks apply when the deformation is one of pure shear or equibiaxial tension. Thus, the introduction of compressibility into the theory does not detract from the correlation with experiment obtained in (I) by use of the incompressible theory. We now show that the simple tension equations ensure that the volume changes must be of the indicated order.

We rewrite equation (15) in the form

$$Jg' = \mu_n(1 - \alpha^{-\frac{1}{2}\alpha_n} J^{\frac{1}{2}\alpha_n})/\lambda, \quad (18)$$

and note that the right-hand side of (18) is typically of order μ/λ . Now, g' is supposed to be a well-behaved function (in particular, we assume it is differentiable); thus, since $g'(1) = 0$ and $g''(1) = 1$, equation (18) implies that J differs from unity by a quantity of order μ/λ in simple tension. This order of volume changes is precisely that recorded in the experimental work of Gee *et al.* (1950), Hewitt & Anthony (1958) and Allen, Bianchi & Price (1963). The same estimate holds in any configuration where one principal component of stress is zero.

(b) Equations for small compressibility

For incompressible elastic solids explicit solution of a number of boundary-value problems is possible. The introduction of compressibility, however, greatly increases the mathematical difficulties involved with the result that few boundary-value problems have been solved; indeed, solution is *possible* only when the strain-energy function is specified and *probable* only when the form of this function is fairly simple. One way of getting round these difficulties to a certain extent is to assume that the material is only slightly compressible and to regard the ratio μ/λ as a small parameter, η say, so that terms of the second and higher orders in η can be neglected. Spencer (1964; see also 1970, § 10) has adopted this approach. He assumed that, when the strain-energy function for an incompressible material is given and the solution to a particular boundary-value problem known, the introduction of a small amount of compressibility can be effected by a small perturbation on the original strain-energy function. Then, by assuming that the displacements, stresses, etc., were correspondingly perturbed, he obtained a solution to a static boundary-value problem, namely the combined extension, inflation and shear of a circular cylindrical tube of slightly compressible isotropic elastic material. Following the method of Spencer, Faulkner (1971) obtained a solution to a problem involving the radial oscillations of a thick spherical shell of the same material.

The specific equations (11), when expanded to the first order in η , also afford the possibility for solution of particular boundary-value problems. Research along these lines is in progress, and the results will be reported in due course. Here, we merely take note of the relevant equations. To the first order in η we write $J = 1 - \eta h$, where h , a function of the deformation, is of order unity. Equations (11) are expanded as

$$\tau_i = \mu_n(\alpha_i^{\alpha_n} - 1) - \mu h + \frac{1}{2}\mu\eta\{g'''(1) + 2\}h^2 \quad (19)$$

to the first order in η , or, in terms of Cauchy stress,

$$\sigma_i = \mu_n(\alpha_i^{\alpha_n} - 1) - \mu h + [\frac{1}{2}\mu g'''(1)h^2 + \mu_n(\alpha_i^{\alpha_n} - 1)h]\eta. \quad (20)$$

The incompressible limit is achieved by letting η tend to zero while keeping $\lambda\eta$ ($= \mu$) fixed. Then the function h is no longer determined by the deformation, and the quantity $\sum_n \mu_n + \mu h$ takes the role of the 'arbitrary' hydrostatic stress.

3. RELATION BETWEEN THEORY AND EXPERIMENT

A number of authors have attempted to obtain theoretical expression for the compressibility data of various materials. Murnaghan (1951, pp. 71–79) used a particular empirical pressure–volume relation in order to fit Bridgman's data for sodium at pressures of up to 10^5 kgf cm^{-2} (10 GPa). This has the form

$$P = K(J^{-k} - 1)/k, \quad (21)$$

where K (a measure of the bulk modulus of the material) and k (≈ 4.0 for sodium) are constants. Equation (21) is a special case of what may be regarded as an extension to macroscopic theory of the 'general' molecular attractive-repulsive force law $P = AJ^{-m} - BJ^{-n}$, where A and B are constants (see, for example, Houwink & de Decker 1971, pp. 12, 13).

Flory (1961), in the context of thermoelasticity, proposed a free-energy function of the form $W = f^* + f$, where f is the 'liquid-like' contribution dependent only on temperature and density (equivalent to the $f(J)$ in this paper for isothermal deformations), and f^* is the contribution due to the elasticity of the molecular network. He specified $f^* = \mu\{\phi(2) - \frac{1}{2} \ln J\}$; because of the factor $\frac{1}{2}$ the ground state is not stress free in this theory unless a compensating term $-\frac{1}{2}\mu \ln J$ is included in the function f (which he did not specify).

Treloar (1969*a, b*), in calculating the effect of volume changes associated with the simple torsion of a rubber cylinder, used, in effect, the correct form $\mu\{\phi(2) - \ln J\}$ of f^* in conjunction with $f = \lambda(J - \ln J - 1)$. The latter term gives a contribution $\lambda(J^{-1} - 1)$ to the pressure; this expression corresponds to $k = 1$ in equation (21) and is linear in the density (or volume) change.

Blatz & Ko (1962) and Blatz (1963) have constructed constitutive laws for compressible continuum and foam rubbers. These laws are based upon the neo-Hookean and Mooney forms of strain-energy function arising in the incompressible theory. In addition, they have used terms of the form (21) in order to fit the data of Bridgman (1944) on the compressibility of rubberlike materials. The agreement between these experimental results and their theory is good, and Ogden (1970*b*) has shown that their strain-energy functions conform with certain inequalities which, from a physical standpoint, one would expect to hold. Levinson & Burgess (1971) have also considered these strain-energy functions in comparison with one of their own construction, this involving a quadratic in the specific volume change $J - 1$.

In Bridgman's experiments a cylindrical plug of material was constrained within a circular cylinder and subject to pressure on its plane face. The configuration may be described as one of uniaxial strain.

Let $a_1 = a$ be the stretch ratio in the direction of compression. Then $a_2 = a_3 = 1$ and $J = a$. From equation (11) the corresponding principal components of Kirchhoff stress are seen to be

$$\left. \begin{aligned} \tau &\equiv \tau_1 = \mu_n(a^{2n} - 1) + \lambda J g', \\ \tau_2 &= \tau_3 = \lambda J g'. \end{aligned} \right\} \quad (22)$$

If P (> 0) is the magnitude of the applied pressure (per unit area) the first of equations (22) gives

$$-aP = \mu_n(a^{2n} - 1) + \lambda ag'(a). \quad (23)$$

The first term on the right-hand side of equation (23) is small by comparison with the second. Thus, neglecting terms of order μ/λ , we may rewrite equation (23) as

$$P = -\lambda g'(a) \quad (24)$$

and, by virtue of equations (22), the configuration approximates to one of all-round hydrostatic pressure. Equation (24) is used here as the basis for comparison with experiment.

It is recalled that the function $F(J)$ (or equivalently $g(J)$) may be written as a linear combination of terms like $(J^\gamma - 1)/\gamma$ (put $\alpha = \beta = 0$ in (4)). For simplicity, and in order to obtain some definite results, we consider g being composed of a single such term. Since we are only considering volume *reductions* here it is appropriate to choose $-\gamma = \beta$, say, where $\beta > 0$. However, it is necessary to include a term corresponding to $\gamma = 0$ so that g conforms with the conditions (9). We then have

$$g = \beta^{-2}\{\beta \ln J + J^{-\beta} - 1\}. \quad (25)$$

It follows from (25) that

$$Jg' = -\beta^{-1}(J^{-\beta} - 1). \quad (26)$$

This differs from the Murnaghan form (21), in effect, by the factor J on the left-hand side. We note that the expression $J(Jg')'$, which arises in connexion with the definition of bulk modulus (see § 4), has the simple, positive-definite form $J^{-\beta}$.

Using (26) in (24), we obtain

$$JP = \lambda \beta^{-1}(J^{-\beta} - 1), \quad (27)$$

while at sufficiently high pressures the further approximation

$$P \approx \lambda \beta^{-1} J^{-(\beta+1)} \quad (28)$$

can be made. Comparison of (28) and the pressure-volume data then enables us to obtain a rough estimate of the values of λ and β required for any given rubberlike solid.

From a logarithmic plot of the (P, J) data it turns out that β lies roughly between 9 and 11 for several rubbers of differing constitution. In fact, for a given rubber, a range of different values of β may be chosen, and it is not easy to determine which value is best. Correspondingly, different values of λ are equally suitable. For demonstration purposes and for ease of calculation we have selected $\beta = 9.0$ so that equation (27) now reads

$$P = \frac{1}{9}\lambda(J^{-10} - J^{-1}). \quad (29)$$

We define the non-dimensional pressure P^* by $P^* = 9P/\lambda$. Equation (29) then gives

$$P^* = J^{-10} - J^{-1}, \quad (30)$$

which does not involve the constant λ . The values of P^* calculated according to equation (30) are plotted against $(1 - J) 10^2$ in the accompanying figure for the range

$J = 1.0$ down to about $J = 0.8$. For comparison with the resulting curve we have plotted data relating to three different rubbers with an appropriately chosen value of λ for each rubber. These are: (i) Koroseal, $\lambda = 2.6 \times 10^4 \text{ kgf cm}^{-2}$ (2.6 GPa); (ii) Goodrich D-402, $\lambda = 3.6 \times 10^4 \text{ kgf cm}^{-2}$ (3.6 GPa) and (iii) rubber 'A', $\lambda = 5.0 \times 10^4 \text{ bars}$. Data for (i) and (ii) are taken from Bridgman's (1944) paper; for these two rubbers the pressures applied were up to $2.5 \times 10^4 \text{ kgf cm}^{-2}$ (2.5 GPa). Data for (iii) are taken from Table 1 of the paper by Adams & Gibson (1930), the origin being adjusted here so that $P = 0$ corresponds to $J = 1$. In this case pressures of up to $1.2 \times 10^4 \text{ bars}$ (1.2 GPa) were applied and measurements taken at more frequent intervals than for (i) and (ii).

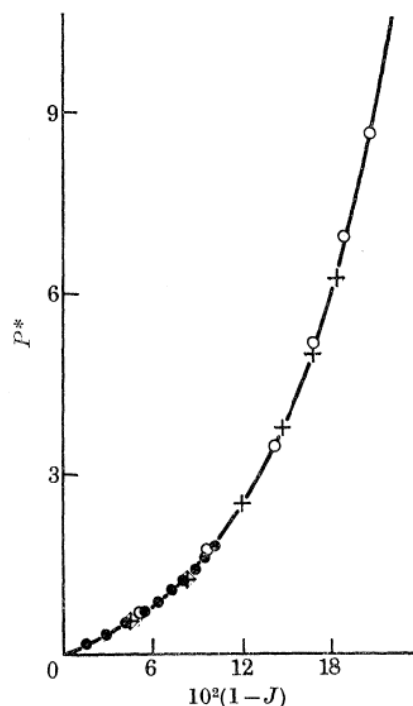


FIGURE 1. Volume ratio–pressure data for the rubbers: (i) $\circ$, (ii) $+$, and (iii) $\bullet$, compared with the curve calculated from the theoretical relation (30).

As can be seen from the figure the agreement between theory and experiment is good for the considered range of pressures, and we conclude that currently available compressibility data can be accounted for by a 'simple' function of the volume ratio.

4. SOME INEQUALITIES ON THE MATERIAL PARAMETERS

The expression 'material parameters' is used here to embrace both material constants and variables (such as the function g) which depend upon the deformation. In ((I), §6) we considered what restrictions it was reasonable to impose on the

material constants α_n and μ_n . The discussion centred on the constitutive inequality of Hill (1968, 1970*b*) which, as far as present knowledge suggests, is in full accord with the mechanical behaviour of elastic solids; it can therefore provisionally be adopted as a means for reducing the class of possible constitutive laws. It was found in (I) that if, for each n , μ_n has the same sign as the corresponding α_n Hill's inequality is satisfied. In general it is not possible to find necessary and sufficient conditions which involve only the constants α_n and μ_n .

Full details of the analysis of this and other inequalities for general isotropic elastic solids have been given by Hill (1970*b*) while Ogden (1970*a, b*) has examined the consequences of the inequality for some particular constitutive laws for compressible isotropic elastic solids. The investigation is continued here in respect of the strain-energy function (10), g being unspecified but subject to the conditions (9) and (13).

Let ϵ represent the Eulerian strain-rate and τ the Kirchhoff stress in symbolic notation, each of ϵ and τ being a vector in a nine-dimensional vector space. Then, for compressible materials, Hill's inequality asserts that the scalar product $(\mathcal{D}\tau/\mathcal{D}t) \epsilon$ is positive definite at all strains and for arbitrary non-zero strain-rates. The notation $\mathcal{D}/\mathcal{D}t$ is used for the rate of change on axes rigidly rotating with the local body spin. The basic tools required in the analysis are given in (I), § 6 and appendix.

From (I), equation (24)) we recall that, if ϵ_{ij} denote specifically the components of the Eulerian strain-rate on the principal axes of the Eulerian† strain ellipsoid (also the principal axes of Cauchy stress for isotropic elastic solids), then

$$\dot{a}_i/a_i = \epsilon_{ii} \quad (i = 1, 2, 3). \quad (31)$$

Also, if we suppose that, relative to the axes of the Eulerian strain ellipsoid, the body spin has non-zero components ω_{ij} ($i \neq j$) then (I), equation (23))

$$-\omega_{ij} = \frac{a_i^2 + a_j^2}{a_i^2 - a_j^2} \epsilon_{ij} \quad (a_i \neq a_j \text{ when } i \neq j). \quad (32)$$

In (31) the superposed dot denotes the material time derivative.

Now suppose that $\hat{\tau}_{ij}$ represent the components of the rigid-body derivative of Kirchhoff stress on the axes of the Eulerian strain ellipsoid. Then, by use of equations (10), (31) and (32), we obtain

$$\hat{\tau}_{ii} \equiv \dot{\tau}_i = \mu_n \alpha_n a_i^{2n} \epsilon_{ii} + \lambda J (Jg')' \epsilon_{kk} \quad (i = 1, 2, 3), \quad (33)$$

$$\text{and} \quad \hat{\tau}_{ij} \equiv -\omega_{ij}(\tau_i - \tau_j) = \mu_n \frac{a_i^{2n} - a_j^{2n}}{a_i^2 - a_j^2} (a_i^2 + a_j^2) \epsilon_{ij} \quad (i \neq j), \quad (34)$$

in each of which the summation is over n only. The scalar product $(\mathcal{D}\tau/\mathcal{D}t) \epsilon$ can then be written explicitly as

$$\hat{\tau}_{ij} \epsilon_{ij} \equiv \mu_n \alpha_n a_i^{2n} \epsilon_{ii}^2 + \mu_n \frac{a_i^{2n} - a_j^{2n}}{a_i^2 - a_j^2} (a_i^2 + a_j^2) \epsilon_{ij}^2 + \lambda J (Jg')' (\epsilon_{kk})^2. \quad (35)$$

In the first and second terms on the right-hand side of (35) there is summation over the repeated indices i, n and i, j, n respectively, while in the third term summation

† This terminology is explained in (I); see also Hill (1970*a, b*).

over k is implied. Each of i, j and k ranges over 1, 2 and 3; but n ranges from 1 to N , where N is unspecified.

When the strain-rate is volume preserving $\epsilon_{kk} = 0$, and the right-hand side of (35) is equal to

$$\mu_n \alpha_n a_i^{2n} \epsilon_{ii}^2 + \mu_n \frac{a_i^{2n} - a_j^{2n}}{a_i^2 - a_j^2} (a_i^2 + a_j^2) \epsilon_{ij}^2. \quad (36)$$

The expression (36) is identical to the corresponding one for incompressible materials ($\epsilon_{kk} \equiv 0$), except that in (36) the product $a_1 a_2 a_3$ is not necessarily equal to unity. It may be remarked that the expression (36) is positive definite if and only if each of the inequalities $\mu_n \alpha_n x^{\alpha_n} > 0$ and $\mu_n (x^{\alpha_n} - y^{\alpha_n})/(x - y) > 0$ holds for all positive x and y . Moreover, it can easily be shown that these two inequalities are equivalent to one another.

As in the incompressible case ((I), equation (26)) it is clear that sufficient conditions for (36) to be strictly positive are

$$\mu_n \alpha_n > 0 \quad (\text{each } n; \text{ no summation}). \quad (37)$$

If there are, say, N (α_n, μ_n) pairs occurring in the strain-energy function and we order $\alpha_1 < \alpha_2 < \dots < \alpha_N$, the conditions $\mu_1 \alpha_1 > 0$ and $\mu_N \alpha_N > 0$ are necessary, but nothing definite can be said about the sign of $\mu_n \alpha_n$ for $1 < n < N$. In general, therefore, we have to be content with sufficient conditions. It was pointed out in (I) that the pairs (α_n, μ_n) considered there in fitting the theory to experimental data do, in fact, conform with the restrictions (37). Ogden & Chadwick (1972) have shown that if the conditions $|\alpha_n| > 1$ (each n) hold then (37) are also sufficient for the existence of a solution to the problem of the axial or torsional shear of a circular cylindrical tube of incompressible isotropic elastic material.

We now return to the expression (35) in order to investigate the sign of $\lambda J(Jg')'$. Since λ is positive it is clear that the conditions (37), together with the restriction

$$(Jg')' \equiv g' + Jg'' > 0, \quad (38)$$

are sufficient for Hill's inequality to hold. Is (38) also a necessary condition? It was indicated in §2 that the restriction $g'' > 0$ might be expected to hold for all J (> 0). Since, according to (13), $g' < 0$ for $J < 1$, it is clear that we must have $g'' > 0$ for $J < 1$ if (38) is to be true for such values of J . For $J > 1$ we have $g' > 0$ from (13) so that (38) then allows the possibility of g'' being negative. In the present context $\lambda \gg \mu$ so the inequality $(\mathcal{D}\tau/\mathcal{D}t)\epsilon > 0$ could be violated in certain configurations if the reverse of (38) were to hold. We conclude that for rubberlike materials the condition (38) is a necessary consequence of Hill's inequality. It should be noted that, if the reverse of (38) were allowed for $J > 1$, the condition $g'' < 0$ would follow by virtue of (13). From a purely *mathematical* viewpoint there is no reason to suppose that (38) should be a necessary consequence of Hill's inequality; for example, this conclusion cannot be drawn if λ and μ are of the same order of magnitude, as they are for many metals in the ground state.

Consider a strain-rate which is a pure dilation ($e_{ij} = \frac{1}{3}e_{kk}\delta_{ij}$). The right-hand side of (35) may then be written as

$$\left\{\frac{1}{3}\mu_n\alpha_n(a_1^{2n} + a_2^{2n} + a_3^{2n}) + 3\lambda J(Jg')'\right\}e_{ij}e_{ij}. \quad (39)$$

When the underlying strain is also a dilation, in which the three stretches have the common value a , we may define the bulk modulus $\kappa(a)$, referred to the deformed state, via the equation

$$a^3\kappa(a) = \frac{1}{3}\mu_n\alpha_n a^{2n} + \lambda J(Jg')'. \quad (40)$$

This follows the definition given by Hill (1970*b*, §7(*a*)) for a general isotropic elastic solid relative to the rigid-body derivative of Kirchhoff stress. According to Hill's inequality the conjunction of (40) and (39) then implies that $\kappa(a) > 0$ for all a . When $a = 1$ we have $\kappa(1) = \lambda + \frac{2}{3}\mu$, as in the classical theory. The bulk modulus may be expressed in terms of density ρ and pressure P as $\rho dP/d\rho - P$. This differs from the conventional definition $dP/d\rho$, though the two are equivalent in the stress-free ground state. When $g'' > 0$ and (38) holds it follows that both $dP/d\rho$ and $\rho dP/d\rho - P$ are positive. If the possibility of g'' being negative is admitted the bulk modulus according to the classical definition will be negative in certain configurations. However, there is no evidence at present to support such a possibility.

In the same fashion a shear modulus $\mu(a)$ may be defined relative to a state of dilatation (see Hill 1970*b*, §7(*a*)) by

$$2a^3\mu(a) = \mu_n\alpha_n a^{2n} \quad (\mu(1) = \mu). \quad (41)$$

In order to create a further correspondence with the classical theory we introduce the variable $\lambda(J)$ defined by

$$\lambda(J)/\lambda \equiv (Jg')' \quad (\lambda(1) = \lambda). \quad (42)$$

Thus, in a state of dilatation, equations (40), (41) and (42) yield the connexion

$$\kappa(a) = \lambda(a^3) + \frac{2}{3}\mu(a), \quad (43)$$

which holds for all a . The inequalities $\mu(a) > 0$ and $\kappa(a) > 0$ result directly from Hill's inequality, and we have asserted that, by virtue of the physical requirement $\lambda \gg \mu$, the restriction $\lambda(J) > 0$ should also follow. And this must certainly be true to within order μ/λ since, according to (43), the condition $\kappa(a) > 0$ leads to $\lambda(a^3)/\lambda$ being greater than a negative quantity of this order.

We now invoke a physical argument which indicates that $\lambda(J)$ should be strictly positive. Consider a cube of material initially of unit edge but currently in a state of dilation with all-round stretch a and associated Kirchhoff stress τ . Suppose a (logarithmic) strain increment is applied in which only one component, e_{33} , is non-zero. The corresponding stress increment $\dot{\tau}_3$ is, according to equation (33),

$$\dot{\tau}_3 = \{\lambda(a^3) + 2\mu(a)\}a^3e_{33}, \quad (44)$$

while the increments in the lateral stresses (proportional to the load increments in this case) are

$$\dot{\tau}_1 = \dot{\tau}_2 = \lambda(a^3)a^3e_{33}. \quad (45)$$

Suppose $a > 1$; then an increase in τ_3 would be required to effect an increase in strain $\epsilon_{33} > 0$ so that $\lambda(a^3) + 2\mu(a) > 0$. This inequality is also a consequence of $\mu(a) > 0$ and $\kappa(a) > 0$. Physically one would expect a resulting tendency to lateral contraction so that compensating increases $\dot{\tau}_1 > 0$ and $\dot{\tau}_2 > 0$ are required to maintain the lateral state of strain. It follows from (45) that $\lambda(a^3) > 0$ in these circumstances. Parallel arguments may be applied in situations where $a < 1$ so that $\lambda(J) > 0$ for all J . From equations (26) and (42) it follows that, for the particular form of g (valid for $J \leq 1$) considered in §3, $\lambda(J) = \lambda J^{-(\beta+1)}$, which is strictly positive.

(a) Other inequalities

The constitutive inequality $(\mathcal{D}\boldsymbol{\tau}/\mathcal{D}t)\boldsymbol{\epsilon} > 0$ is one of a number of inequalities examined by Hill (1970b) in respect of isotropic elastic solids. In particular he considered the class of inequalities $\boldsymbol{s}\boldsymbol{\epsilon} > 0$ where, in symbolic notation, $\boldsymbol{s}$ is any stress-rate (referred to the current configuration) given by

$$J\boldsymbol{s} = (\mathcal{D}\boldsymbol{\tau}/\mathcal{D}t) - m(\boldsymbol{\tau} \times \boldsymbol{\epsilon} + \boldsymbol{\epsilon} \times \boldsymbol{\tau}),$$

m being a (real) constant, and a cross denoting the tensor inner product. Hill (1968) has shown that these inequalities may be violated in certain configurations, unless $m = 0$, for all incompressible elastic materials. For a general isotropic elastic solid we have (Hill 1970b, equation (18))

$$J\boldsymbol{s}\boldsymbol{\epsilon} = \left(a_j \frac{\partial \tau_i}{\partial a_j} - 2m\tau_i \delta_{ij}\right) \epsilon_{ii} \epsilon_{jj} + \left(\frac{a_i^2 + a_j^2}{a_i^2 - a_j^2}(\tau_i - \tau_j) - m(\tau_i + \tau_j)\right) \epsilon_{ij}^2, \quad (46)$$

where summation over i and j from 1 to 3 is implied. In respect of the strain-energy function (10) the inequality $\boldsymbol{s}\boldsymbol{\epsilon} > 0$ requires, among other conditions, that

$$\mu_n \frac{a_i^{2n} - a_j^{2n}}{a_i^2 - a_j^2} (a_i^2 + a_j^2) - m\mu_n (a_i^{2n} + a_j^{2n} - 2) - \lambda m J g' > 0. \quad (47)$$

Now, from (13), $g' \geq 0$ according as $J \geq 1$, and on physical grounds, we know that $\lambda \gg \mu$. Therefore, for a hydrostatic state of stress the inequality (47) can be violated for $m > 0$ (< 0) by a hydrostatic tension (pressure) of sufficient magnitude. If $m = 0$ the inequality holds by virtue of the conditions (37).

When $m = \frac{1}{2}$ the inequality $\boldsymbol{s}\boldsymbol{\epsilon} > 0$ is the strengthened form of the hypothesis of Coleman & Noll (1964, equation 6.18; further details may be found in Truesdell & Noll 1965, §§52, 87). Although Coleman & Noll state (1964, p. 89) that '*the consequences of the TI (thermostatic inequality) for finite elastic deformations seem to be in accord with all experience in solids*', the above argument clearly contradicts this statement and the Coleman–Noll hypothesis must finally be rejected, at least in respect of rubberlike solids.

In order for (47) to be violated when $m > 0$ a state of all-round tension (or something near it) is required. This would seem to be difficult to achieve in practice. However, we now propose an experiment in which an approximate state of all-round tension could be attained.

Take a sample of rubberlike material in the shape of a circular disk whose thickness is small compared with the radius. Suppose this is bonded between two circular plates of a tension apparatus, the plates being of equal area to the test piece. Let equal and opposite tensions τ (per unit area) be applied normal to each plate. In the region between the central parts of the two plates the configuration will effectively be one of uniform uniaxial extension, there being no corresponding lateral contraction except in the edge region.

Equations (22) are therefore applicable to this situation if we take $a > 1$. When the tension τ (> 0) is applied the volume will increase in the centre region by the factor $J = a$, and g' must be positive. The lateral stresses $\tau_2 = \tau_3 = \lambda J g'$ are also positive and serve to maintain the lateral state of strain. Since μ/λ is a small quantity $\tau_1 \approx \lambda J g'$ and the state of stress approximates to all-round tension in the central region. The results of such an experiment would enable us to obtain a specific expression for g holding for $J > 1$.

(b) *A further consequence of the inequalities*

Consider once again the simple tension equation (15) (or (18)) relating the stretch a to the density ratio J . For a homogeneous specimen under this homogeneous deformation J is also the volume ratio V/V_0 , where V_0 and V are the initial and current volumes of the specimen. Here we investigate whether the inequalities (37) and (38) imply a volume expansion or contraction when the specimen is extended. Let δa and δJ denote respectively an increase in the extension ratio a and the change induced in J . From equation (15) we obtain

$$\{\lambda J(Jg')' + \frac{1}{2}\mu_n \alpha_n J^{\frac{1}{2}\alpha_n} a^{-\frac{1}{2}\alpha_n}\} \frac{\delta J}{J} = \frac{1}{2}\mu_n \alpha_n J^{\frac{1}{2}\alpha_n} a^{-\frac{1}{2}\alpha_n} \frac{\delta a}{a}. \quad (48)$$

When the inequalities (37) and (38) hold equation (48) asserts that an *increase* in the length of the test specimen is always accompanied by an increase in volume, this increase being of order μ/λ . A similar conclusion may be drawn for pure shear and equibiaxial tension. In the ground state $a = J = 1$ and $\delta J = \mu \delta a / (\lambda + \mu)$, as in the classical theory.

This prediction corresponds very well with the experimental evidence at stretches where crystallization is not significant. Gee *et al.* (1950), Hewitt & Anthony (1958) and Allen *et al.* (1963) have shown specific volume increases of the order 10^{-4} which is of the same order as μ/λ . After about 400 % extension, however, the volume decreases below its initial value. This phenomenon is due to crystallization and cannot therefore be accounted for in terms of the present theory, and certainly not by means of equation (48) as long as the inequalities (37) and (38) hold. Most situations can be approximated by a strain-energy function with a single (α_n, μ_n) term, (α_1, μ_1) say in addition to the g term. Then the restriction $\mu_1 \alpha_1 > 0$ is an essential requirement, and the inequality (38) in conjunction with this provides a pair of necessary and sufficient conditions for Hill's inequality to hold. In these circumstances (48) can never predict a volume contraction.

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